

Obliczyć całkę $\Leftrightarrow$ znaleźć pole obszaru ograniczonego
 osi Ox funkcją $y = \frac{\ln x}{x^3}$
 i prostymi $x=1$, $x=3$.

$$\begin{aligned} \int_1^3 \frac{\ln x}{x^3} dx &= \left\{ \begin{array}{l} f = \ln x \quad f' = \frac{1}{x} \\ g' = \frac{1}{x^3} \quad g = -\frac{1}{2x^2} \end{array} \right\} = -\frac{\ln x}{2x^2} \Big|_1^3 + \frac{1}{2} \int_1^3 \frac{1}{x^3} dx = \\ &= -\frac{\ln 3}{18} + \frac{1}{2} \left(-\frac{1}{2x^2} \Big|_1^3 \right) = -\frac{\ln 3}{18} - \frac{1}{4} \left(\frac{1}{x^2} \Big|_1^3 \right) = \\ &= -\frac{\ln 3}{18} - \frac{1}{4} \left(\frac{1}{9} - 1 \right) \end{aligned}$$

$$\int_0^1 \frac{x}{25+49x^4} dx = \left\{ \begin{array}{l} 7x^2 \Big|_0^1 = t \Big|_0^7 \\ 14x dx = dt \\ \frac{x dx}{25+49x^4} = \frac{\frac{1}{14} dt}{25+t^2} \end{array} \right\} = \frac{1}{14} \int_0^7 \frac{1}{t^2+25} dt$$

$$\begin{aligned} &= \frac{1}{14} \int_0^7 \frac{1}{25(\frac{t^2}{25}+1)} dt = \frac{1}{14 \cdot 25} \int_0^7 \frac{1}{\frac{t^2}{25}+1} dt = \\ &= \left\{ \begin{array}{l} u \Big|_0^{\frac{7}{5}} = \frac{t}{5} \Big|_0^7 \\ du = \frac{1}{5} dt \end{array} \right\} = \frac{1}{14 \cdot 25} \cdot \int_0^{\frac{7}{5}} \frac{5 du}{1+u^2} du = \\ &= \frac{1}{70} \arctg \frac{7}{5} \end{aligned}$$

$$\int_0^1 \frac{1}{e^x + 1} dx = \left\{ \begin{array}{l} t \Big|_0^1 = e^x \Big|_1^e \\ dt = e^x dx \\ \frac{dx}{e^x + 1} = \frac{dt}{t} \cdot \frac{1}{t+1} \end{array} \right\} = \int_1^e \frac{1}{t(t+1)} dt =$$

$$= \int_1^e \left(\frac{1}{t} - \frac{1}{t+1} \right) dt =$$

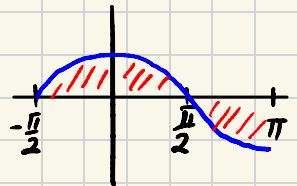
$$\frac{1}{t(t+1)} = \frac{A}{t} + \frac{B}{t+1} = \begin{array}{l} A+B=0 \\ A=1 \\ B=-1 \end{array} = \int_1^e \frac{1}{t} dt - \int_1^e \frac{1}{t+1} dt =$$

$$= \frac{A(t+1) + Bt}{t(t+1)} = \frac{(A+B)t + A}{t(t+1)} = \ln|t| \Big|_1^e - \ln|t+1| \Big|_1^e$$

$$\int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx = \begin{cases} \frac{(n-1)!!}{n!!} & n \text{ nieparzyste} \\ \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & n \text{ parzyste} \end{cases}$$

$n = 1, 2, \dots$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^5 x dx = \int_0^{\frac{\pi}{2}} \cos^5 x dx = \frac{4!!}{5!!} = \frac{4 \cdot 2}{5 \cdot 3} = \frac{8}{15}$$



$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^6 x dx = 5 \cdot \frac{5!!}{6!!} \cdot \frac{\pi}{2} = \frac{5 \cdot 5 \cdot 3}{6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{25\pi}{32}$$

$$\int_0^2 |x^2 - 3x + 2| dx =$$



$$= \int_0^1 x^2 - 3x + 2 dx - \int_1^2 x^2 - 3x + 2 dx =$$

$$= \int_0^1 x^2 dx - \int_0^1 3x dx + \int_0^1 2 dx - \int_1^2 x^2 dx + \int_1^2 3x dx - \int_1^2 2 dx =$$

$$= \left. \frac{1}{3}x^3 \right|_0^1 - \left. \frac{3}{2}x^2 \right|_0^1 + 2x \Big|_0^1 - \left. \frac{1}{3}x^3 \right|_1^2 + \left. \frac{3}{2}x^2 \right|_1^2 - 2x \Big|_1^2 =$$

$$= \frac{1}{3} - \frac{3}{2} + 2 - \left(\frac{8}{3} - \frac{1}{3} \right) + \left(6 - \frac{3}{2} \right) - (4 - 2) =$$

$$\int_{-\infty}^0 x \cdot e^{2x} dx = \lim_{A \rightarrow -\infty} \int_A^0 x \cdot e^{2x} dx = \lim_{A \rightarrow -\infty} \left(\frac{e^{2x}}{2} x - \frac{1}{4} e^{2x} \right) \Big|_A^0 =$$

$$= \lim_{A \rightarrow -\infty} \left(-\frac{1}{4} - \left(\frac{e^{2A} \cdot A}{2} - \frac{e^{2A}}{2} \right) \right) = -\frac{1}{4} - \lim_{A \rightarrow -\infty} \left(\frac{e^{2A} \cdot A}{2} - \frac{e^{2A}}{2} \right)$$

$$\int x \cdot e^{2x} dx = \left\{ \begin{array}{l} f = x \quad f' = 1 \\ g' = e^{2x} \quad g = \frac{1}{2} e^{2x} \end{array} \right\} =$$

$$= \frac{1}{2} e^{2x} \cdot x - \frac{1}{2} \int e^{2x} dx = \frac{e^{2x}}{2} x - \frac{1}{4} e^{2x}$$

$$\int_1^{\infty} \frac{1}{x^3+x} dx = \int_1^{\infty} \frac{1}{x} dx - \frac{1}{2} \int_1^{\infty} \frac{2x}{x^2+1} dx =$$

$$\frac{1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} \quad = \lim_{B \rightarrow \infty} (\ln|x|) \Big|_1^B - \frac{1}{2} \lim_{B \rightarrow \infty} (\ln(x^2+1)) \Big|_1^B =$$

$$1 = Ax^2 + A + Bx^2 + Cx$$

$$A+B=0 \quad B=-1$$

$$C=0$$

$$A=1$$

$$= \lim_{B \rightarrow \infty} (\ln B) - \frac{1}{2} \lim_{B \rightarrow \infty} (\ln(B^2+1) - \ln 2)$$

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rozbieżna

$$\int_3^6 \frac{1}{x^2-7x+10} dx = \int_3^5 \frac{1}{(x-2)(x-5)} dx$$

Skonwertujemy na
następujących ćwiczeniach